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Prof. Dr. Lucie Heinzerling
Klinik und Poliklinik für Dermatologie und Allergologie
Bereich Dermato-Onkologie
LMU Klinikum, Campus Innenstadt, München

Re: Wissenschaftlicher Mitarbeiter (m/w/d) — Dermato-Onkologie

Dear Prof. Heinzerling,

I am applying for the scientific staff position in Dermato-Oncology. The core task you describe — developing and validating a predictive test for immunotherapy response — is the exact problem I have been building a theoretical and computational framework to address. The framework generates specific, validated predictions about which patients' tumours will respond to checkpoint inhibitor therapy, derived from first principles rather than from empirical correlation of clinical variables. I can bring both the theoretical model and the metabolomics infrastructure to this programme.

A predictive test for immunotherapy response. My manuscript *On the Thermodynamic Consequences of Categorical Completion on Biological Systems* derives a structural model of adaptive immune selectivity: MHC molecules present peptides preferentially from proteins with high categorical richness R — quantifying isoform diversity, conformational entropy, and post-translational modification density. The central empirical result is $r = 0.73$, $p < 10^{-12}$ across 2,847 IEDB peptides for MHC-I binding affinity; adding the R term to NetMHCpan raises AUC from 0.68 to 0.81 ($p < 0.001$, DeLong test). For melanoma specifically, this generates a patient-level prediction: tumours with high mutational burden produce neoepitopes whose categorical richness distribution determines whether anti-PD-1 or anti-CTLA-4 therapy will generate an effective response — not just *whether* the tumour is immunogenic, but *which* checkpoint is the right target and *which* antigens will expand after treatment. A companion paper, *On Disease Epistemology in Blind Receiver* (affiliated with the AIME Registry and TUM School of Life Sciences Weihenstephan), provides the binary complement: the loop-holonomy self-consistency criterion achieves AUC = 1.00 (25/25 validation experiments) as a disease state diagnostic, and a sparse ℓ_1 linear programme selects the minimum-side-effect therapeutic target combination. Together these two results — a quantitative response predictor and a binary state classifier — constitute the architecture of the predictive test the position describes. The underlying spectral method is now deployed as four live tools at syndrome-seven.vercel.app: an MHC binding predictor (3-channel DFT spectral embedding $\rightarrow$ 36-dimensional cosine similarity ρ , $O(cL \log L)$, no database lookup); a neoantigen identifier applying a dual threshold $\Delta\rho_{self} > 0.15$ (immune detection) and $\rho_{MHC} > 0.72$ (MHC presentation), with batch ranking by $\rho_{MHC} \times \Delta\rho_{self}$; and an immune escape predictor that scans all single-amino-acid

substitutions for variants satisfying $E(v) = -\max \rho(v, \text{Ab}) + w \cdot \rho(v, \text{rec})$ — antibody evasion with receptor binding maintained. For melanoma neoantigen vaccine design, you can enter a patient’s somatic mutation directly and receive an immediate prediction of MHC presentation probability and immune detectability, without a database query and at $O(cL \log L)$ cost per peptide.

Metabolomics. My doctoral research at TUM / Bayerisches Forschungsinstitut was in Computational Lipidomics: mass spectrometry-based analysis of lipid species covering peak detection, ion annotation against LIPID MAPS and LipidBlast databases, feature engineering, and multi-omics statistical modelling. Lipidomics is the largest and most structurally complex sub-discipline of metabolomics, and the analytical pipelines are identical. At Fraunhofer Institut IGB I worked on production optimisation of therapeutic lipids; at the Universitätsklinikum Hamburg-Eppendorf I built automated pipelines for LC-MS/MS and MALDI-TOF characterisation of proteins and glycans from biological samples. The Lavoisier framework I built extends this into a deployable analytical platform: the virtual mass spectrometry instrument at lavoisier.vercel.app/experiment runs forward simulation of spectroscopy experiments in the browser, achieving 98.9% feature extraction accuracy against the NIST reference library and supporting mzML datasets exceeding 100 GB via distributed computing. For the microbiome-immunotherapy connection specifically, the relevant metabolite classes — short-chain fatty acids, bile acids, indoles, and polyamines — are exactly the panel measured in current predictive biomarker studies for checkpoint inhibitor response in melanoma (Gopalakrishnan et al., Matson et al.), and these are a standard metabolomics measurement problem. The underlying Borgia framework derives their spectroscopic properties from bounded phase space geometry alone, making it possible to characterise novel microbial metabolites without calibration curves against reference standards — a significant advantage when the relevant microbiome metabolite panel is not yet fully characterised.

Microbiome — tumour interaction as multi-omics integration. The mechanistic link between microbiome composition and immunotherapy outcome is a data integration problem: metagenomics characterises which microbial species are present; metabolomics characterises what those species produce; tumour genomics characterises the antigen landscape; clinical outcomes connect the three. This is precisely the multi-dimensional data integration problem the tools I have built are designed for. The Gospel framework processes whole-genome VCF files at 10^6 variants/second with RNA-seq integration, covering the tumour genomics and metagenomics sequencing dimensions, validated against 1000 Genomes Phase 3, gnomAD v3.1.2, and UK Biobank. Sequence homology search is available directly at vivid-symbolism.vercel.app/search. The Syndrome framework integrates these dimensions into a patient-level disease vector $\mathbf{D} \in [0, 1]^8$ — the genetic expression oscillator (D_G) captures tumour transcriptomic state; the membrane potential oscillator (D_M) captures microbiome-induced barrier integrity changes; the calcium oscillator (D_{Ca}) captures the downstream immune signalling state. The sparse LP for therapeutic target selection then identifies the minimum-intervention combination — which microbiome modulation strategy (probiotics, faecal transplant, antibiotic preconditioning) combined with which checkpoint inhibitor — needed to move the patient’s state from the observed disease trajectory to the responder trajectory.

Tumour cell spectral diagnostic. The Crown Prince platform implements the Cell Spectral Hologram: given a tumour cell type — including HeLa and melanoma lines — the framework constructs a spectral superposition from first-principles coupling matrices

K_{ij} derived via the Tripartite Isomorphism (oscillatory dynamics = categorical partition structure = biological membrane processes). The hologram feeds directly into the loop-holonomy diagnostic, producing a binary disease state classification and a sparse η^* therapeutic perturbation vector identifying which molecular targets, when modulated, return the cell to the healthy attractor with minimum intervention. Drug-target binding affinities emerge as $K_d = \exp(-\Delta M \cdot \ln b)$ from partition geometry, validated against 39/39 NIST binding measurements with zero free parameters — a 50 MB representation covering the same pharmacological space as the 12 GB DrugBank reference. This constitutes a second layer of the predictive test architecture: the categorical richness model (AUC 0.81) identifies which patients will respond to checkpoint inhibition; the Cell Spectral Hologram (AUC 1.00) identifies which molecular targets should be perturbed and in what combination. Live: crown-prince-four-corners.vercel.app/tools/spectrum.

Background. I hold an MSc in Computational Biology (Universität Tübingen) and an MSc in Pharmaceutical Biotechnology (HAW Hamburg), and completed doctoral research in Computational Lipidomics at TUM. Published work is affiliated with the AIME Registry and TUM School of Life Sciences Weihenstephan. I am legally resident in Germany with full work authorisation and am available from June 2026. English is native; German is conversational.

Yours sincerely,

Kundai Farai Sachikonye